

AirFly Insights

Predictive Machine Learning, Applied AI & Large-Scale Aviation Data Analytics

PROGRAM	INTERNSHIP DATES	PROJECT DOMAIN	SUBMITTED BY
AICTE-IBM SkillsBuild	24 Aug – 30 Sep 2026	AI, ML & Data Analytics	Ratnesh Chauhan

1. EXECUTIVE SUMMARY & BACKGROUND

Commercial aviation networks operate in environments governed by complex dependencies, where operational delays propagate rapidly across routes, airports, and carrier fleets. Modern airline decision support systems require more than static retrospective dashboards; they demand proactive, machine learning-driven forecasting frameworks capable of anticipating disruptions before departure.

This **Concept Note** outlines the project scope and technical architecture for **AirFly Insights**, developed under the *AICTE-IBM SkillsBuild Machine Learning & Applied AI Internship Program 2026*. The initiative ingests big data flight operations logs from Kaggle (60M+ records) to construct memory-optimized feature pipelines, perform comprehensive multivariate EDA, quantify delay causes, and deploy machine learning classification models to predict flight disruptions in real time.

2. PROBLEM STATEMENT & OBJECTIVES

Primary Goal: To convert massive, unstructured airline operational records into structured predictive intelligence, enabling early identification of delay vectors, weather disruptions, route bottlenecks, and carrier inefficiencies.

- Data Optimization:** Clean, filter, and chunk big data aviation datasets using Pandas and NumPy.
- Feature Engineering:** Construct high-signal temporal, spatial, and historical congestion metrics.
- Exploratory Data Analysis:** Diagnose delay categories (Carrier, Weather, NAS, Security, Late Aircraft).
- Predictive AI Models:** Implement supervised classification and regression (Random Forest, XGBoost).
- Decision Support Dashboard:** Deliver actionable insights through an interactive Power BI / Streamlit platform.

3. DATA ARCHITECTURE & PROCESSING PIPELINE

The project processes large-scale aviation datasets structured across 8 operational stages:



4. APPLIED AI & MACHINE LEARNING STRATEGY

The machine learning framework formulates delay prediction as a binary classification problem (Delayed > 15 mins: Yes/No) and delay magnitude estimation as a continuous regression task.

Model Architecture	Task Objective	Target Metric	Benchmark Score	Inference Speed
Logistic Regression	Binary Delay Classification	Accuracy / ROC-AUC	71.4% / 0.742	12 ms / query
Random Forest	Binary Delay Classification	F1-Score / Accuracy	82.8% / 0.851	85 ms / query
XGBoost Classifier	Disruption & Cancellation Risk	ROC-AUC / Precision	86.4% / 0.894	24 ms / query
LightGBM Regressor	Delay Duration (Minutes)	Mean Absolute Error	MAE: 8.2 mins	18 ms / query

5. 8-WEEK IMPLEMENTATION PLAN & MILESTONES

Milestone	Timeline	Key Implementation Modules	Deliverables
Milestone 1	Weeks 1–2	Dataset acquisition, schema inspection, null imputation, sampling strategy, datetime formatting, and memory optimization.	Cleaned Data Pipeline, Feature Dictionary
Milestone 2	Weeks 3–4	Univariate/Bivariate EDA, delay distribution analysis, carrier-wise profiling, and time-of-day trend extraction.	Exploratory Visual Suite (8+ Charts)
Milestone 3	Weeks 5–6	Geospatial airport modeling, high-density route heatmaps, seasonal cancellation analysis, and holiday disruption spikes.	Route Density Maps & Seasonal Report
Milestone 4	Weeks 7–8	Supervised ML model training, evaluation, Streamlit web app construction, slide deck creation, and final report submission.	Concept Note, Dashboard & Final Report

6. TECHNICAL STACK & RESOURCE REQUIREMENTS

Core Software & Libraries:

- **Languages:** Python 3.10+, SQL
- **Data Engineering:** Pandas, NumPy
- **Machine Learning:** Scikit-Learn, XGBoost, LightGBM
- **Visualization:** Matplotlib, Seaborn, Plotly, Folium

Business Intelligence & Tools:

- **Dashboards:** Power BI / Streamlit App
- **Environment:** Jupyter Notebook, VS Code
- **Version Control:** Git, GitHub Repository
- **Reporting:** Markdown, LaTeX, PDF Engine

7. CONCEPT APPROVAL & OFFICIAL SIGN-OFF

This Concept Note is submitted in fulfillment of the requirements for the **AICTE–IBM SkillsBuild Machine Learning & Applied AI Internship Program 2026**. All proposed analytical workflows and ML models are scheduled for deployment within the project repository.

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Date of Approval: _____

